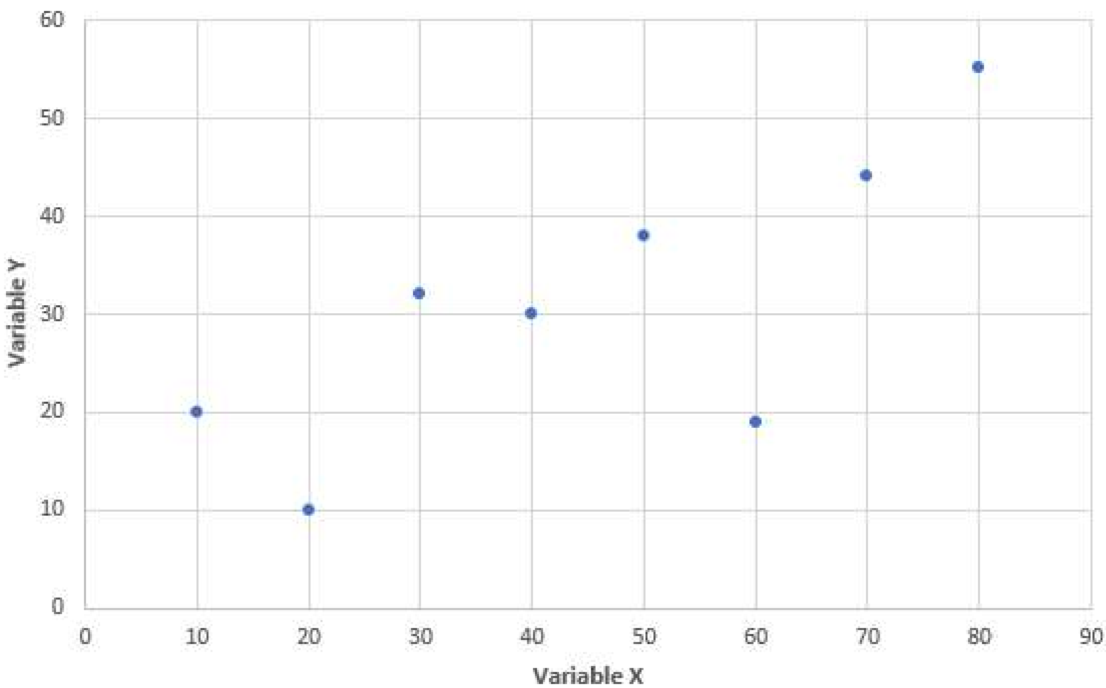
Data Analytics

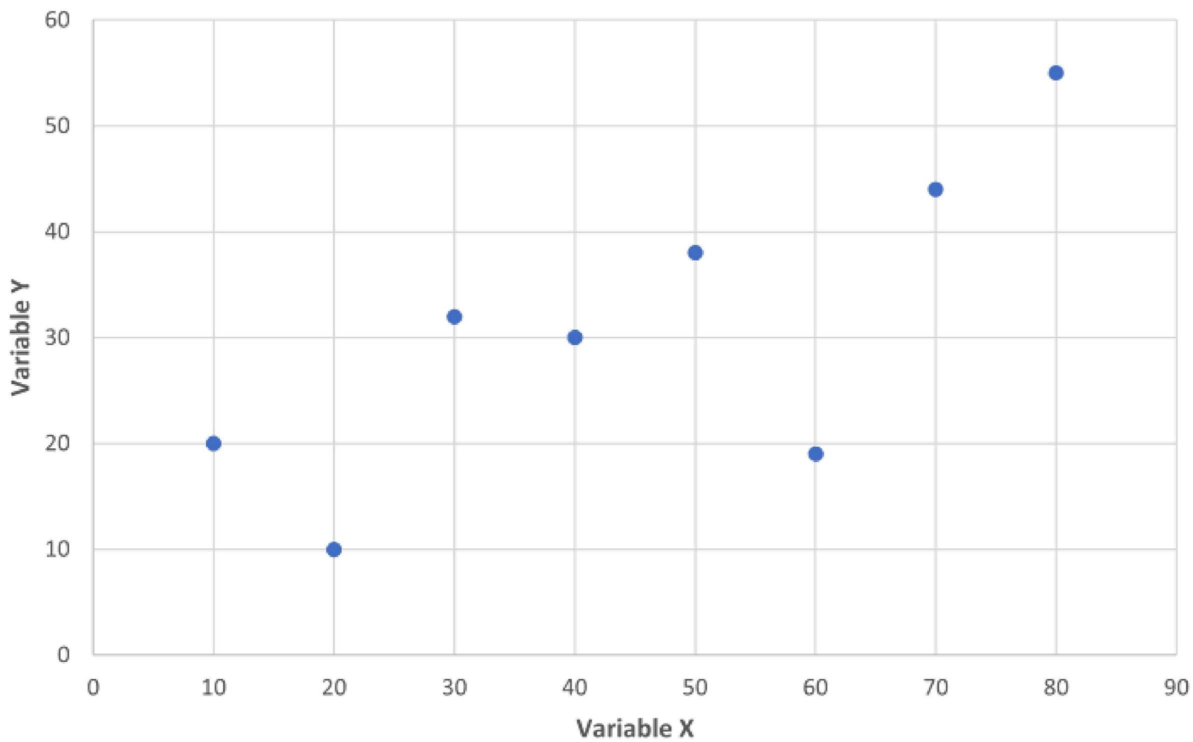
* Indicates required question

1) Visualization that displays the correlation between variable X and Y
•Answer the below questions by selecting the correct option from each drop-down list.



1. a) What is the direction of correlation between variable X and variable Y *

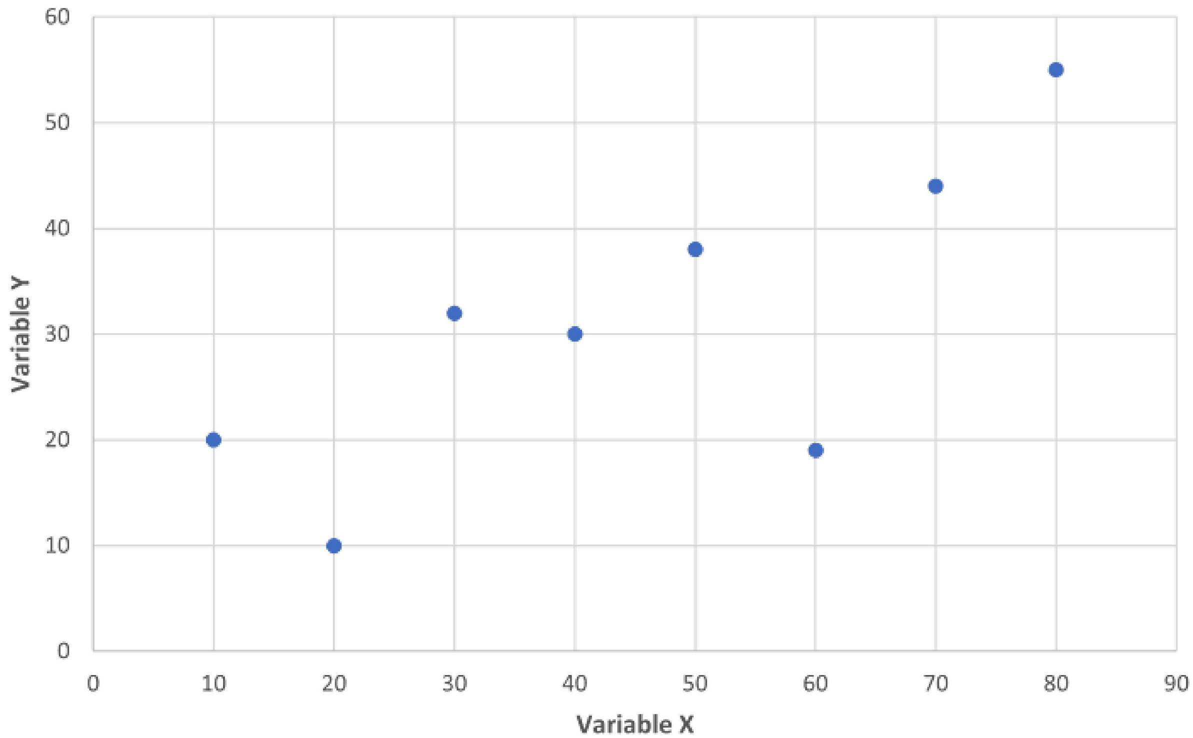
Dropdown 1 point



Mark only one oval.

- Positive
- Negative
- Zero

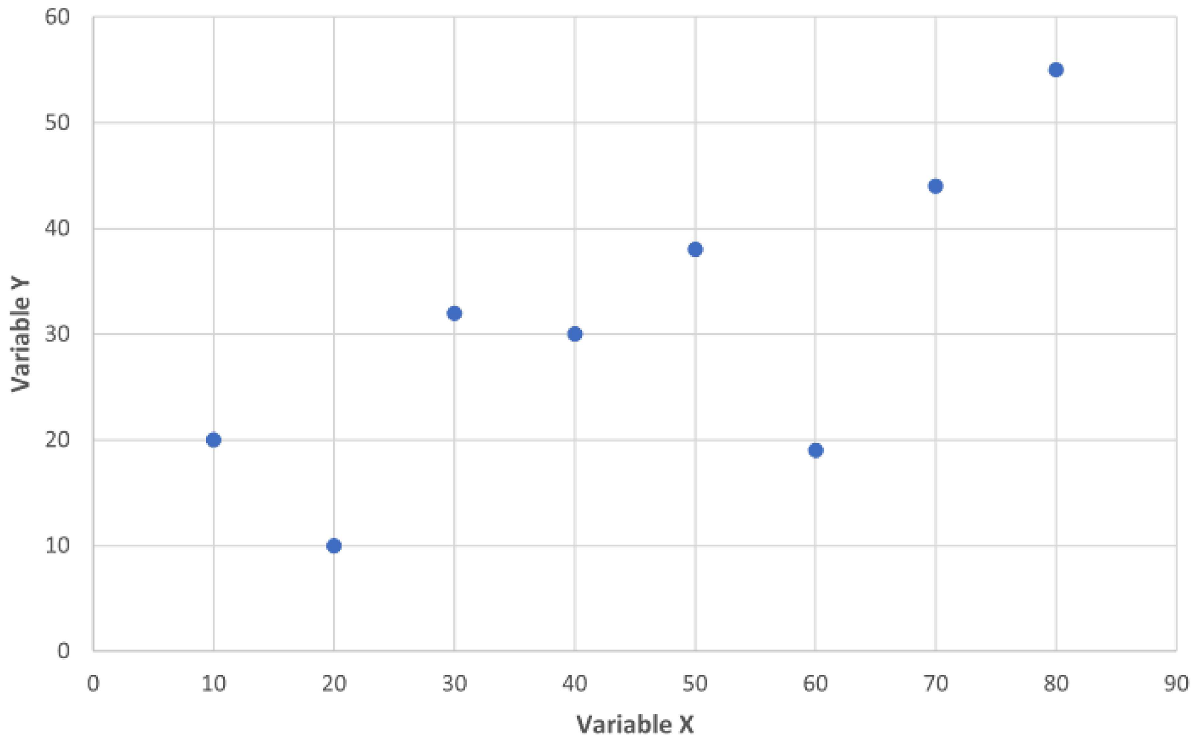
2. b) Which correlation range most likely describe relationship between the variable X and Y * Dropdown 1 point



Mark only one oval.

- No correlation(r=0.00)
- Some correlation(0.00<r<0.99)
- Perfect correlation(r=1.00)

3. c) An analyst claims the visualization implies the variable X causes variable Y.is the analyst correct in the assertion * 1 point



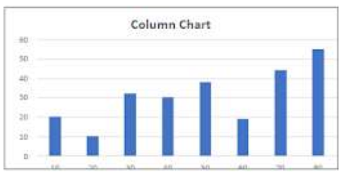
Mark only one oval.

- Yes
- No

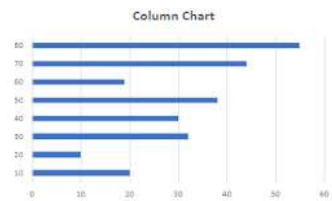
4. 2) Which visualization type is commonly used to display the distribution of a continuous variable, with variable values on the x axis and corresponding frequencies on the Y axis?
Select the correct visualization type in the answer area.

1 point

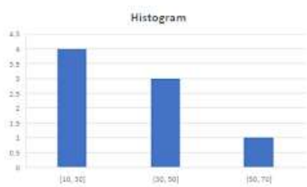
Mark only one oval.



Option A



Option B



Option C

Option D

5. 3)What data structure describes the following data? *
["Aabid", "Jesenia", "Mark"]

Dropdown

1 point

Mark only one oval.

- List
- Multi-dimensional
- Table
- Graph

6. 4) You need to create a data view based on aggregations for further visual analysis. Your data includes sales information for the past five years for food products at your company's stores. Each product belongs to one category. For example milk belongs to dairy category. * 2 points

The data view must meet the following requirements
•Include all products and their associated categories
•Include sales sub-total for each category and year
•Display grand total of sales for each category
•Create a summary of each category for every year

Which two aggregation methods should you use to create the data view? (Choose 2)

Note:- you will get a partial credit for each correct selection

Tick all that apply.

- [ ] Pivoting
- [ ] Filtering
- [ ] Merging
- [ ] Grouping

7. 5) You are performing descriptive analysis on quarterly sales data
Move the appropriate statistical metrics from the list on the left to the correct locations on the right. You may use each metric once, more than once
Note:-partial credits will be given for each correct response

4 points

Statistical Metrics

a) Average
b) Max
C) Median
D) Min
E) Mode
F) Sum

Region	Quarter 1	Quarter 2	Quarter 3	Quarter 4
North	25,000	30,000	40,000	50,000
South	35,000	45,000	40,000	55,000
East	35,000	32,500	41,000	52,500
West	34,500	30,000	42,500	55000
1	129,500	137,500	163,500	212,500
2	35,000	45,000	42,500	55,000
3	25,000	30,000	40,000	50,000
4	35,000	30,000	40,000	55,000

Mark only one oval per row.

	1	2	3	4
Average	◯	◯	◯	◯
Max	◯	◯	◯	◯
Median	◯	◯	◯	◯
Min	◯	◯	◯	◯
Mode	◯	◯	◯	◯
Sum	◯	◯	◯	◯

8. 6) Which data type can store a phrase or sentence * 1 point

Mark only one oval.

- Integer
- Character
- String
- Boolean

9. 7) A coworker is having trouble joining two database tables, Table A and Table B, that were imported from CSV files. They say the tables have no common values. * 2 points

You need to troubleshoot the problem. You look at the data in the original CSV file and find that the RowKey values in the TableA file and the RowID values in the TableB file look identical. Both have three numbers followed by a dash(-) and two letters

Which two actions should you complete next? (choose 2)

Note:- you will receive partial credit for the correct selection

Tick all that apply.

- [ ] Verify that the data in the database was imported as a numeric data type
- [ ] Trim empty spaces from both of the valid characters
- [ ] Visually compare the database values to the CSV values
- [ ] Trim empty spaces from only the right side of the valid characters

10. 8) A file named courses data contains the following content * 1 point

Title|Number|Hours
Algebra|MT101|3
History|HS201|3
Physics|PS302|4
Music|MS101|2
Art|AR201|2

You need to use Python to read the data from the file so that you can import it in to database file

11. 9) Each month you need to automatically transform the data from two XML documents into a single flat file with columns and rows that excel can open and interpret. The document names and structure remain constant. You know the relationship between the two XML documents * 2 points

Which two resources can you use?(choose 2)

note:- You will receive partial credit for each correct selection

Tick all that apply.

- [ ] Json
- [ ] Power Query for Excel (M)
- [ ] Microsoft Excel
- [ ] Python

12. 10) What is the goal of data privacy and protection laws such as GDPR, FERPA and HIPAA? * 1 point

Mark only one oval.

- To hold violators accountable for mishandling data
- To tax companies that use private data
- To ensure that companies openly share industry data
- To protect companies from liability related to private data

13. 11) You have a small dataset that contains personally identifiable information (PII). You need to provide the data to an outside source for additional processing * 1 point

What could you do to protect the PII but still allow you to eventually relate the additional analysis to your original data?

Mark only one oval.

- Employ pseudonymization on the PII and the pseudonym as the key between the new and original datasets
- Retain every text based PII in the original dataset but convert them to number-based features in the new dataset
- Randomly shuffle the original dataset so that each given piece of PII is no longer associated with a particular user
- Remove every instance of PII in the original dataset and add them back after the new dataset is retrieved

14. 12) The visualization and the data table depict housing prices in a region * for each statement about the visualization, select True or False
Note:- you will receive partial credit for each correct selection

3 points

year	Price
2016	$ 1,50,000
2017	$ 1,75,000
2018	$ 2,00,000
2019	$ 2,25,000
2020	$ 2,50,000
2021	$ 2,75,000
2022	$ 3,00,000
2023	$ 3,25,000
2024	$ 3,50,000
2025	$ 4,25,000

Mark only one oval per row.

	True	False
The visualization uses scaling manipulation		
An annual increase of $25,000 occurs between 2016 and 2025		
The visualization accurately depicts the housing prices shown		

15. 13) What is a raw data * Dropdown 1 point

Mark only one oval.

- Unprocessed Data
- Purely numerical Data
- Summarized Data
- Visualized Data

16. 14) You are analyzing sales activity that occurs on national holidays 0 points

What level of data granularity will enable you to perform the most precise analysis?

Data Granularity Options

Mark only one oval.

- Years
- Months
- Weeks
- Days
- Hours

17. 15) What is an example of data cleaning? * 1 point

Mark only one oval.

- Ensuring that the data in word table uses a consistent font
- Adding quotation marks to the beginning and end of a tab-delimited file
- Removing non-printable characters from a comma-delimited file
- Arranging Excel data rows in an order that is easy for a user or read

18. 16) A popular social media site records and counts clicks, likes, and dislikes and other user interactions * 1 point

What type of data is collected?

Mark only one oval.

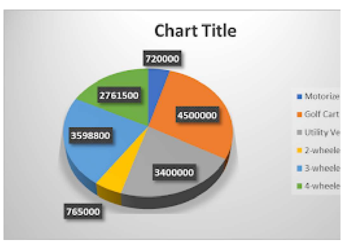
- A. Continuous data
- B. Qualitative data
- C. Imputed data
- D. Big data

19. 17) You work for a recreational sports company. The tables shows the companies recreational vehicle sales
You need to show how each vehicle type contribute to the company's total sales
Which visualization should you use? Select the correct visualization in the answer area

* 1 point

Vehicle Type	Quantity Sold	Price	Total
Motorized bike	600	1,200	720,000
Golf cart	600	7,500	4,500,000
Utility Vehicle	425	8000	3,400,000
2-wheel Scooter	850	900	765,000
3-Wheel Scooter	1,200	2,999	3,598,800
4-wheel Scooter	789	3,500	2,761,500
Total	**4,464**	**24,099**	**15,745,300**

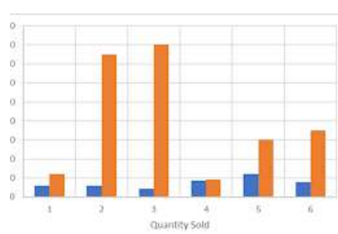
Mark only one oval.



Option 1

Option 2

Option 3



Option 4

20. 18) From the data in the table below, you create a pivot table to show the combined number of certified virtual and in-person teachers for each class at each school. * 4 points

Move the appropriate labels from the list on the left to the correct locations in the Pivot tables on the right. You may use each label once or not at all

School	Class	Format	Certified teacher
School A	Neworking	In Person	6
School A	Networking	Virtual	5
School A	Data Analytics	In Person	2
School A	Data Analytics	Virtual	3
School B	Networking	In Person	9
School B	Networking	Virtual	7
School B	Data Analytics	In Person	2
School B	Data Analytics	Virtual	4

Labels

- Data Analytics
- Networking
- In-Person
- Virtual
- School A
- School B

	Label 1	Label 2
Label 3	11	5
Label 4	16	6

Mark only one oval per row.

	Data Analytics	Networking	In-Person	Virtue	School A	School B
Label 1						
Label 2						
Label 3						
Label 4						

21. 19) Your company has summarized a large data set for the region you live in. You need to compare results from urban and rural communities within your region

What is the fastest way to obtain the information

* 1 point

Mark only one oval.

- A. Collect a new Data Sample
- B. Review data from neighbouring regions
- C. Disaggregate the data
- D. Aggregate the data

22. 20) What concept allows analysts to drill down into data and examine different levels of information that may be crucial in diagnostic analytics

* 1 point

Mark only one oval.

- Granularity
- Completeness
- Interpretability
- Transparency

23. 21) For each statement about the machine learning select True or False * 3 points

Note You will receive partial credit for each correct selection

Answer Area

Mark only one oval per row.

	True	False
Machine learning can predict the probability of Rain in a region by examing known weather patterns	⬭	⬭
Machine learning can help determine whether a candidate will pass an exam without looking at historical scores	⬭	⬭
Machine Learning can be used to automatically decline financial purchases based on previous purchase activity	⬭	⬭

24. 22) Which Data structure have multiple rows and columns? * 0 points

Mark only one oval.

- ⬭ Series
- ⬭ Table
- ⬭ One-dimensional Array
- ⬭ List

25. 23) In which scenario will artificial intelligence (AI) provide the greatest benefit? * 1 point

Mark only one oval.

- Predicting maintenance requirements for a international rental car company's fleet vehicles
- Determining the statistical mean, mode, and standard deviation of the grades for a class
- Recording daily sales for the three stores owned by one franchise owner
- Interpreting fundraising sales data for a college soccer team

26. 24) Move each function from the list on the left to the correct description on the right * 4 points

Tick all that apply.

	Count()	Max()	Min()	Sum()
Returns the largest value	☐	☐	☐	☐
Returns the smallest value	☐	☐	☐	☐
Returns the number of Values	☐	☐	☐	☐
Returns the total of the values	☐	☐	☐	☐

27. 25) Person A has 5 coins and person B has 10 coins * 1 point

Which type of data does the number of coins represents?

Mark only one oval.

- Ordinal Data
- Metadata
- Qualitative data
- Quantitative data

28. 26) You have a dataset that includes product review scores and demographic information about the reviewers. There are no subcategories associated with the demographic answers. The table shows a selection of the data. * 1 point
Which Scenario is an example of disaggregating the dataset

Product	Review Score	Review id	Industry	Ethnicity
AX-150	74	123	Education	Asian
BK-330	82	124	Finance	Latino or Hispanic
BK-315	79	125	Health Care	Native Hawaiian or Pacific Islander
CX-290	86	126	Other	African-American
BD-250	61	127	Finance	Other
CD-140	35	128	Food Services	Caucasian
AX-310	84	129	Eductaion	Caucasian

Mark only one oval.

- By average and mode of the scores for each product grouped by the ethnicity of the reviewers
- Display the overall average and mode of all scores on a per-products basis
- Display a list of ethnicities that are included in the other option
- Display the overall average and mode of all scores and a count of all reviews

29. 27) Which two concepts are commonly associated with artificial intelligence (AI) in data analytics? (Choose 2) * 2 points

Tick all that apply.

- [ ] Cost- Benefit Analysis
- [ ] Stakeholder Mapping
- [ ] Automation
- [ ] Machine Learning

30. 28) For which two reasons is it risky to make generalizations from limited sample data? (choose 2) * 2 points

Tick all that apply.

- [ ] Limited data Samples are easier to collect
- [ ] A limited sample may not represent a larger population
- [ ] Findings from a smaller sample size may not be as precise
- [ ] Analyzing data from a smaller sample size is faster

31. 29) You are reviewing a database of restaurant menu items. The table below shows a selection of the data. * 1 point

You need to display only items on the desert menu with a type of cake

What should you do to nondestructively limit the data display?

Item	Type	Menu	Gluten-free	Vegan
Croque Monsieur	Sandwich	Lunch	Optional	No
French Onion Grilled Cheese	Sandwich	Lunch	Optional	No
Lemon Meringue Pie	Pie	Dessert	No	No
Matcha Slice	Cake	Dessert	No	No
Shrimp and crab Louie	Salad	Lunch; Dinner	Yes	No
Spinach-Articoke Panini Bites	Sandwich	Lunch	Optional	Yes
Strawberry Pecan Chicken Delight	Salad	Lunch; Dinner	Yes	Optional
Stuffed Portobello	Appetizer	Dinnner	Optional	Yes
Vegan Chocolate Cherry Torte	Cake	Dessert	Yes	Yes

Mark only one oval.

- Group the data by menu and then group the data on the desert menu by type
- Delete all data that has a menu other than desert. Then delete all data that has a type other than cake
- Add two slicers, one for menu and one for type. Set the menu slicer to desert and the type slicer to cake
- Sort the data by menu and within each menu, Sort by type

32. 30) You want to show a friend your monthly budget breakdown to prove that most of your expenditure is food costs. You create a table that shows the flow of money as it moves one budget category to the next.
Which visualization type should you use to display your analysis based on the table shown? * 1 point

Source Category	Spend category	Amount
Total Monthly Budget	Total Monthly Budget	1,000
Total Monthly Budget	Gym Membership	100
Total Monthly budget	Rent	200
Total Monthly Budget	Food and Entertainment	700
Food and Entertainment	Food Costs	400
Food and Entertainment	Entertainment Costs	300
Entertainment Costs	Movie Theater and Plays	200
Enterainment COsts	Other	100

Mark only one oval.

- Time Series Chart
- Correlation Chart
- Sankey Chart
- Classification Chart

33. 31) The Marketing team wants to know which market segment have the highest sales in the last year * 1 point

Which type of data analysis should they use?

Mark only one oval.

- Perspective analytics
- Diagnostic Analytics
- Predictive Analytics
- Descriptive Analytics

34. 32)You Believe Playing video game's increases the chance of man getting heart attack. In your research you notice equal evidences in favouring your hypothesis and opposed to it. You tried hours trying to identify the problems with the evidence opposed to your hypothesis, but readily accept the evidence in favor

* 1 point

Which type of bias are you demonstrating?

Mark only one oval.

- Motivated Reasoning
- Confirmation Bias
- Anchoring Bias
- Sampling Bias

35. 33) For each statement About the data disaggregation, select True or False

* 3 points

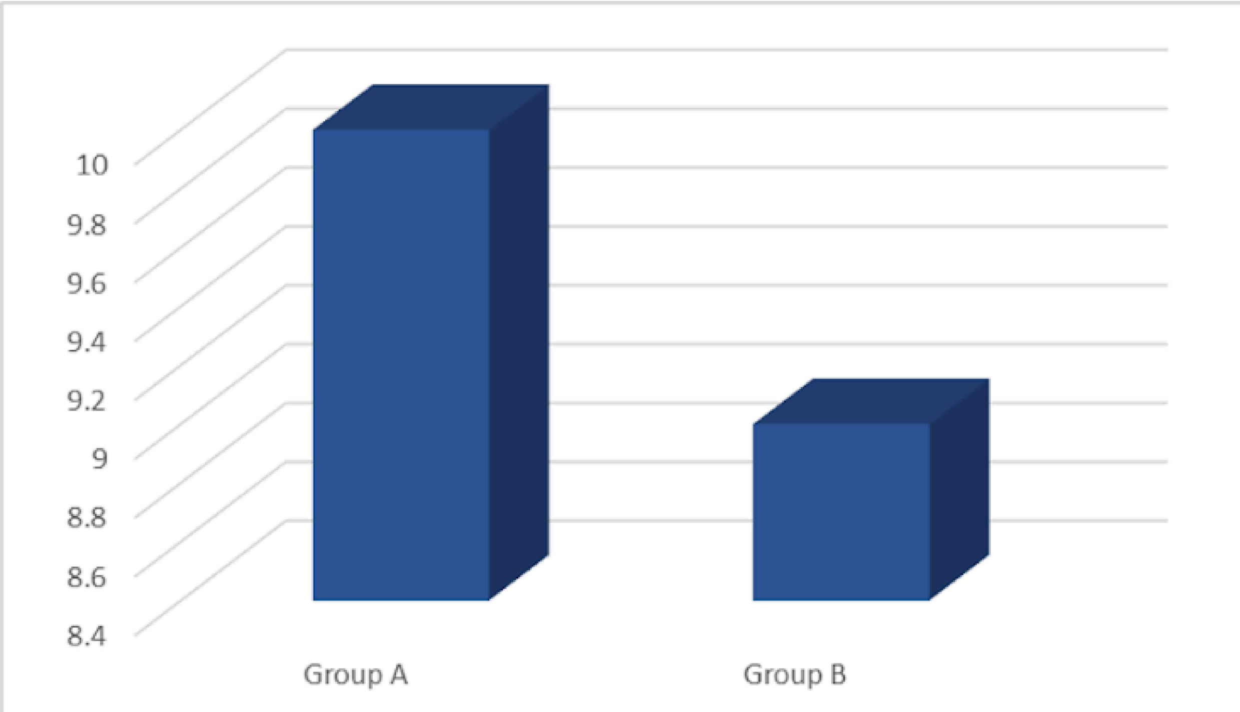
Note:- You will receive partial credits for the correct options choose

Mark only one oval per row.

	True	False
Data disaggregation provides a summary of the data		
Data disaggregation combines data from different sources		
Data Disaggregation can clarify trends and patterns among subgroups		

36. 34) A college shows you the chart below to indicate that group A has performed significantly better than group B on a recent assignment. You don't know the sample size and the result of the statistical testing. * 1 point

Which chart element creates the impression of a significant score difference?



Mark only one oval.

- The X-axis unit of Measurement
- The Y-Axis unit of measurement
- The Z-Axis Unit of Measurement
- The Color differentiation

37. 35) You conduct a study to identify how much people exercise daily. You recruit all the study participants at the gyms * 1 point

Which type of bias are you demonstrating?

Mark only one oval.

- Anchoring bias
- Confirmation Bias
- Motivated Bias
- Sampling bias

38. 36) Which Statement correctly assigns a string to the variable that is name score? * 1 point

Mark only one oval.

- Score=true
- Score=String[7]
- Score="&"
- Score= 7

39. 37) You are using data analytics to help answer business questions about a new product your company released * 4 points
Move each type of data analytics from the list on the left to the correct question on the right
Note:- you will receive partial credit for each correct match

Tick all that apply.

	Why did it happens?	What action should we take next?	What might happen in future?	What happened in the initial product release?
Descriptive Analysis	☐	☐	☐	☐
Diagnostic Analysis	☐	☐	☐	☐
Predictive Analysis	☐	☐	☐	☐
Perspective Analysis	☐	☐	☐	☐

40. 38) You run a t-test with alpha value of 5% (a= 0.05) in order to test an alternative hypothesis (H1). You finish the analysis and discover the P-value is 0.017 * 1 point

What can you conclude about the null hypothesis (H0)?

Mark only one oval.

- You modify the null hypothesis (H0)
- You accept the null hypothesis (H0)
- You fail to reject the null Hypothesis (H0)
- You reject the null hypothesis (H0)

41. 39) You have a comma-delimited file with 100,000 rows and 200 columns of phone sales data. One column represents the Phone manufacturer. * 1 point

You need to analyze all sales data for a specific manufacturer

Which technique should you use?

Mark only one oval.

- Deleting
- Transposing
- Truncating
- Filtering

42. 40) What is Meta-data * 1 point

Mark only one oval.

- The text content of a message
- A context that gives data mining
- Numerical facts
- Statistics